

AN2DL - First Challenge Report

Backtothefeatures

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1 Introduction

This project investigates the performance of **deep learning** techniques for time-series classification [9].

- The objective is to **classify** user-based sequential data from the *Pirate Pain* dataset, which contains **multivariate temporal features** to determine each pirate's level of pain.
- The work involves **data preprocessing, normalization, and user-based stratified splitting** to ensure fair evaluation across unseen users and the selection of models best suited for the task.

2 Analysis

2.1 Dataset Characteristics

The dataset consists of time-series recordings collected from 661 distinct pirates. Each individual contributed 160 temporal samples, capturing the evolution of 31 joint angles and four pain survey scores. The target variable represents the perceived level of pain (*no pain*, *low pain*, *high pain*).

2.2 Main Challenges

Several challenges were encountered during model development [4]:

Dataset imbalance The distribution of the three pain classes is not uniform (Figure 1), possibly leading to bias toward majority categories.

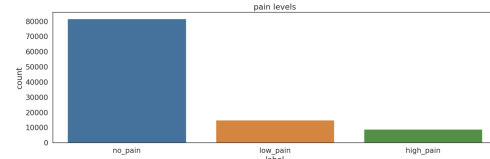


Figure 1: Number of labels per class

Low number of samples Each subject provided only one temporal sequence, limiting variability and increasing overfitting risk.

High number of features The large number of variables increases the risk of spurious correlations specific to the training set.

2.3 Initial Assumptions

Before developing the models, several assumptions guided the design:

- **Categorical attributes** (e.g., number of legs, eyes, hands) were converted to numerical representations.
- **Feature normalization** using Min-Max scaling was applied to stabilize training.

- A **user-based split** between training and validation sets prevented information leakage.
- For the **Transformer** and **L-SFAN**, each training window represents an entire sequence, as they are better at extracting long-term temporal and spatial context. For the **GRU**, exploiting short-range temporal patterns, the sequences were instead windowed.

3 Method

3.1 Models

Three neural architectures were evaluated to capture temporal and spatial aspects of pain-related movement.

GRU. The **Gated Recurrent Unit (GRU)**[2] network modeled sequential dependencies in behavioral time-series. The best configuration used two layers, hidden size $H = 128$, and dropout 0.1. The bidirectional GRU concatenated forward and backward hidden states for classification into three pain levels. The long continuous input was transformed via windowing and striding into many short, partially overlapping time sequences to improve the GRU ability to learn temporal dynamics efficiently. Furthermore, Cross-Entropy Loss [6] was weighted to account for class imbalance, and a Reduce-on-Plateau [1] strategy was implemented.

Transformer.[11] **PiratePainTransformer** leveraged self-attention to capture both local and long-range dependencies. Four encoder layers ($d_{model} = 128$, $n_{heads} = 8$) used GELU activation and residual connections. An attention pooling layer aggregated temporal information:

$$\alpha_t = \frac{e^{\mathbf{w}^\top \mathbf{h}_t}}{\sum_j e^{\mathbf{w}^\top \mathbf{h}_j}}, \quad \mathbf{z} = \sum_t \alpha_t \mathbf{h}_t,$$

allowing the model to emphasize time steps most indicative of pain.

L-SFAN.[10] The **PiratePain-L-SFAN** adapted the Lightweight Spatially-Focused Attention Network for multivariate motion data. Two convolutional blocks (Conv-BN-ReLU-Pool) extracted spatio-temporal features, followed by Temporal Average Pooling (TAP). A multi-head attention mechanism ($H = 5$) modeled dependencies among joints:

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V},$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ are query, key, and value projections of the feature maps. This allows the model to focus on coordinated joint activations typical of pain-related movements, emphasizing spatial reasoning over temporal order. Seeking to obtain stronger results, a 5-Fold Cross Validation was implemented on top of the L-SFAN to better assess the model’s stability. This ensures that performance gains reflect true generalization rather than a favorable split.

3.2 Data Augmentation

To address the strong class imbalance and increase data diversity, two complementary augmentation strategies were adopted.

Manual augmentation perturbed existing sequences using small Gaussian jitter [5] applied only to dynamic features, limited temporal shifts to preserve motion continuity, and edge padding to avoid introducing artificial discontinuities, providing controlled variability in the minority classes.

GAN-based augmentation employed a conditional Wasserstein GAN (WGAN) specifically designed for multivariate time series. The generator produced new sequences conditioned on the target pain label, while the critic enforced realistic temporal dynamics through Wasserstein-based training. Careful training was required to avoid mode collapse.

4 Experiments

Different settings and hyperparameters have been evaluated. The results of the best models on the Kaggle dataset, along with the results on the validation split are presented in the following table:

Table 1: Comparison of best-performing models.

Model	Kaggle F1	Val. Acc.	Val. F1
BiGRU baseline	0.8885	0.8788	0.7230
Transformer	0.8212	0.9092	0.8307
L-SFAN	0.9135	0.9596	0.9241
GAN L-SFAN	0.9144	0.9394	0.8897
GAN Transformer	0.9146	0.9192	0.8562
Man. L-SFAN	0.9156	0.9293	0.8891
Ensemble L-SFAN	0.9244	-	-
5-Fold-CV L-SFAN	0.9091	0.9323	0.8354
BiGRU Revisited	0.9365	0.9269	0.9261

5 Results

As can be seen in Table 1, the **L-SFAN** achieved the highest overall accuracy (95.96%) and macro F1 (0.92) on the validation set, outperforming the **GRU baseline** (87.88%, 0.72). Both **GAN** and **manually Augmented L-SFANs** maintained high accuracy (93.94% and 92.93%), while furtherly specialising respectively in the *low pain* and *high pain* classes. Therefore, for the final submission, an **ensemble of the three L-SFANs** was developed to leverage their complementary strengths. The **Transformer** models, while achieving strong results with around 90% accuracy, consistently underperformed, which can be explained both by the need of larger datasets [12] to learn effective attention patterns and by their focus on modelling long-range temporal dependencies [3], whereas in this dataset the discriminative information is largely spatial (the relative activation patterns across joints and pain features). However, the **windowing** strategy applied to the GRU proved crucial: by creating diverse training samples from the same subject while preserving local temporal dynamics, it effectively regularized the model and captured fine-grained pain transitions that full-sequence models missed, allowing it to achieve the highest F1 score on Kaggle.

6 Discussion

6.1 Strengths

The pipeline evaluates both **temporal-focused** models (Transformers, GRU) and **spatial-focused** models (L-SFAN), offering complementary perspectives on motion-related pain patterns. The results highlight that, in small clinical datasets, the ability of GRUs to model short-term neuromuscular dynamics without relying on long-range dependencies can yield superior generalization [7]. The **windowing approach** proved doubly beneficial in enabling the model to learn from local temporal contexts rather than overfitting to full-sequence patterns. Both **augmentation strategies** effectively generated class-specific samples, reducing imbalance without additional data.

6.2 Limitations

Despite augmentation, **class imbalance** remains challenging. GAN-generated samples did not always

capture the full variability of genuine pain expressions and a coplex architecture was need to avoid mode collapse. Training multiple models increased **computational cost**, and the ensemble **reduced interpretability** compared to single models.

6.3 Methodological and Clinical Insights

Normalization ensured that all features contributed on the same scale, while several regularization strategies, including **early stopping**, **dropout** and **label smooting**, were applied to prevent overfitting and help the models focus on meaningful temporal structure. Clinically, improved high-pain discrimination enables timely interventions. The spatial-temporal modeling aligns with the clinical understanding of pain as both localized tension and time-evolving motion.

7 Conclusions

This work proposed a multi-architecture approach for automatic pain recognition, combining attention-based models such as **L-SFAN** and **Transformer** with the **GRU**, which, with windowing and striding, demonstrated superior generalization capabilities. This outcome reveals the intertwining of model architecture and data preprocessing. Rather than processing entire sequences, it learned robust local temporal patterns from overlapping windows, effectively combining data augmentation with architectural simplicity. On the other hand, the L-SFAN highlighted the importance of spatial attention over joint features is more effective than purely temporal models in this setting. An **ensemble approach**, combining three specialized L-SFAN models, each optimized respectively for *no pain*, *low pain*, and *high pain* cases, furtherly improved robustness. To cope with the limited and imbalanced dataset, both **manual augmentation** (jitter + temporal shifts) and a **conditional GAN** were introduced.

Future Directions

Future work will focus on **time-series-specific augmentation methods**, including **TimeGAN** [14], to preserve temporal dynamics and on improving **explainability** via visualization of key temporal segments to support clinical interpretation.

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